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GUEST EDITORIAL

Synthesizing traditional biogeography with microbial ecology: the importance of dormancy

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ABSTRACT

The discovery of biogeographical patterns among microbial communities has led to a focus on the empirical evaluation of the importance of dispersal limitation in microbial biota. As a result, the spatial distribution of microbial diversity has been increasingly studied while the synthesis of biogeographical theory with microbial ecology remains undeveloped. To make biogeographical theory relevant to microbial ecology, microbial traits that potentially affect the distribution of microbial diversity need to be considered. Given that many microorganisms in natural environments are in a state of dormancy and that dormancy is an important microbial fitness trait, I provide a first attempt to account for the effects of dormancy on microbial biogeography by treating dormancy as a fundamental biogeographical response. I discuss the effects of dormancy on the equilibrium theory of island biogeography and on the unified neutral theory of biodiversity and biogeography, and suggest how the equilibrium theory of island biogeography can produce predictions approaching those of the Baas-Becking hypothesis (i.e. everything is everywhere, but the environment selects). In addition, I present a conceptual model of the unified neutral theory of biodiversity and biogeography, generalized to account for dormancy, from which a full model can be constructed for species with or without dormant life history stages.

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Keywords

Dormancy, ecological neutral theory, equilibrium theory, island biogeography, microbial biogeography, microbial ecology.

Most, if not all, species are limited in their capacity for dispersal. Dispersal barriers prevent each species from establishing a non-porous global range. These are assumptions of dispersal limitation that affect the fundamental biogeographical unit, the species geographical range (Lomolino *et al.*, 2006). The historical evidence of the effects of dispersal limitation on the ability of species to track geographical changes in environmental conditions and maintain connectivity between populations is present in the distribution of extant and extinct biodiversity. As an impetus for speciation and extinction, dispersal limitation contributes to the evolutionary context of biogeography.

In contrast, a predominant hypothesis in microbial biogeography suggests that dispersal limitation is unimportant in microbial systems. The widely known Baas-Becking hypothesis (BBH) states that 'everything is everywhere, but the environment selects'. The BBH asserts that populations of free-living microorganisms with highly resilient life stages can overcome dispersal limitation (Baas-Becking, 1934; de Wit & Bouvier, 2006). If members of a species could be dispersed everywhere, their absence from a location would be the result of a failure to maintain viability. Thus, if environmental selection resulting in local extinction did not occur, everything would be everywhere. If correct, the BBH renders the study of microbial biogeography a matter of understanding the species niche, denying the possibility for observing historical effects of dispersal limitation. It is no surprise, then, that a major focus of microbial biogeography has been to evaluate the importance of dispersal limitation in structuring microbial systems (Staley & Gosink, 1999; Whitaker *et al.*, 2003; Hughes Martiny *et al.*, 2006; Woodcock *et al.*, 2006; Jenkins *et al.*, 2007; Hooper *et al.*, 2008; Östman *et al.*, 2009).

Despite acceptance for the last 75 years, the BBH has been challenged when biogeographical patterns such as taxa–area relationships (Green *et al.*, 2004; Horner-Devine *et al.*, 2004; Noguez *et al.*, 2005) and the effects of geographical isolation (Papke *et al.*, 2003; Whitaker *et al.*, 2003) have been reported. However, multiple factors have prevented a comprehensive

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rejection of the BBH. These include support for the BBH (Fenchel & Finlay, 2004; Hooper *et al.*, 2008), difficulties of separating geographical variation resulting from dispersal limitation from that resulting from environmental selection (Fenchel & Finlay, 2005; Fierer, 2008), and the common failure to detect rare taxa (Woodcock *et al.*, 2006). While few reject the BBH based on a literal interpretation, some regard the BBH as a metaphor (Foissner, 2006; Weisse, 2006). Indeed, if the BBH were a metaphor, the point would be that microorganisms generally have unparalleled capacities to expand or maintain their ranges. A task for biogeography is to evaluate whether such capacities characterize a different biogeography.

The historical justification for unhindered dispersal has been that microbial capacities for dispersal are affected by small individual masses and immense population sizes (Fenchel & Finlay, 2004; Hughes Martiny *et al.*, 2006; de Wit & Bouvier, 2006). Microscopic masses allow individuals to be dispersed via wind and ocean current, and to be distributed by larger organisms. Microbial populations can be immense, and an extremely low individual probability of dispersing continental-scale distances translates to a reasonable probability for the species. While passive dispersal does not necessarily result in a cosmopolitan distribution, and the positive effect of microscopic size on dispersal distance has been questioned (Foissner, 2006; Jenkins *et al.*, 2007), these effects have been considered to be important. However, most research has not focused on two crucial assumptions of the BBH (de Wit & Bouvier, 2006) that are widely accepted. First, many free-living microorganisms in aquatic and terrestrial environments are dormant (Stevenson, 1978; Luna *et al.*, 2002; Prosser *et al.*, 2007). In fact, < 10% of bacterial cells within soil and aquatic environments are typically active (Gasol *et al.*, 1995; del Giorgio *et al.*, 1997; Luna *et al.*, 2002). Second, dormant microbial populations may persist at nearly undetectable densities, an observation echoed among island biogeographical studies of plant seed banks (e.g. Whittaker *et al.*, 1995, 2000) and recent microbial biogeographical studies (Woodcock *et al.*, 2006; Hubert *et al.*, 2009).

Forms of dormancy exhibited by microorganisms include physical forms such as spores and cysts, but also physiological responses that reduce metabolic rate and prevent reproduction. States of dormancy range from readily reactivated to viable but non-culturable, and from metabolically down-regulated to cryptobiotic. Dormancy, as a general property of microorganisms (Postgate, 2000), has become a central interest in microbial ecology (Roszak & Colwell, 1987; Prosser *et al.*, 2007; Colwell, 2009), and it has been suggested that dormancy is the single most important fitness trait explaining the distribution and persistence of bacteria in aquatic systems (Stevenson, 1978). Because dormancy has proven to be important to the study of microbial ecology, it is likely that dormancy has important influences on microbial biogeography and should be explicitly included in its development. Here, I expand on previous work by showing how dormancy can be given greater consideration in the theoretical development of microbial biogeography.

DORMANCY: A FUNDAMENTAL BIOGEOGRAPHICAL RESPONSE

There are three fundamental biogeographical responses recognized in traditional biogeography that affect the species geographical range (Lomolino *et al.*, 2006). Species can respond to a changing environment by evolving novel adaptations, by shifting their range through dispersal, or by going locally, regionally or globally extinct. Given the widespread capacity and occurrence of dormancy among microorganisms, we should consider whether dormancy may be an alternative to extinction when genetic adaptation and dispersal fail to preserve the lineage. Microbial dispersal is presumed to be more likely with the aid of dormancy; however, dispersal is passive, potentially slow, and will not necessarily lead to suitable areas. While some individuals may be dispersed great distances to more suitable habitats, remaining dormant individuals could persist until suitable conditions for growth and reproduction return. Although populations that do not disperse can track environmental changes by producing more adapted offspring, genetic adaptation may take several generations. Dormancy is an immediate response of individuals that may allow populations to track environmental changes more closely. With regard to the ability to persist in or disperse through adverse environments in dormant stages, the species range may show greater resistance to collapse and exhibit the capacity to expand across otherwise unsuitable environments. With dormancy framed as a fundamental biogeographical response, we can begin to discuss how dormancy might interact with other responses to affect the biogeography of microorganisms.

RECONCILING THE BBH WITH TRADITIONAL BIOGEOGRAPHY

Although the BBH is considered to be a niche assembly hypothesis, the prediction that the environment selects is rarely, if ever, challenged. In fact, the BBH relies heavily upon dispersal and resistance to extinction, and it is in respect of these processes that it draws contention. Recognizing this, I will explore the BBH as a special case of a general biogeographical phenomenon, the maintenance of species richness resulting from the countervailing processes of immigration and extinction. I will begin by using dormancy to reconcile the BBH with a classic theory of dispersal assembly of island communities, the equilibrium theory of island biogeography (MacArthur & Wilson, 1967). Although dormancy has not been previously developed as a driving mechanism within island biogeography, some authors have recognized the potential effects of dormancy on persistence and turnover among island communities (e.g. Holt, 1992; Whittaker *et al.*, 1995, 2000).

THE EQUILIBRIUM THEORY OF ISLAND BIOGEOGRAPHY

MacArthur & Wilson (1967) showed how island communities may be assembled by immigration and maintained at

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equilibrium by extinction. As species from a mainland colonize an island, new species will be represented less frequently among immigrants. If species differ in dispersal abilities, the decreasing pool of potential immigrants will become increasingly composed of those species less capable of dispersal. Hence, the rate of species immigration decreases more quickly as island species richness increases (Fig 1a). As the number of species on an island increases, more species become available to go extinct and the overall rate of extinction increases. The later a species arrives, the more it competes with established species for increasingly limited resources. Hence, the overall rate of extinction increases more quickly as island species richness increases (Fig. 1a). Eventually, immigration and extinction reach similar rates and an equilibrium number of species is reached with a rate of turnover (Fig. 1a). The basic form of the equilibrium theory predicts that immigration rates are affected by the distance from a mainland source to an island, whereby a decrease in distance allows more species to immigrate. Rates of extinction are affected by island area, whereby small islands tend to support smaller populations. Assuming that extinction includes a stochastic component, smaller populations are more likely to go extinct through random drift. Here, I will use the graphical representation of the theory as a framework upon which to discuss the effects of dormancy.

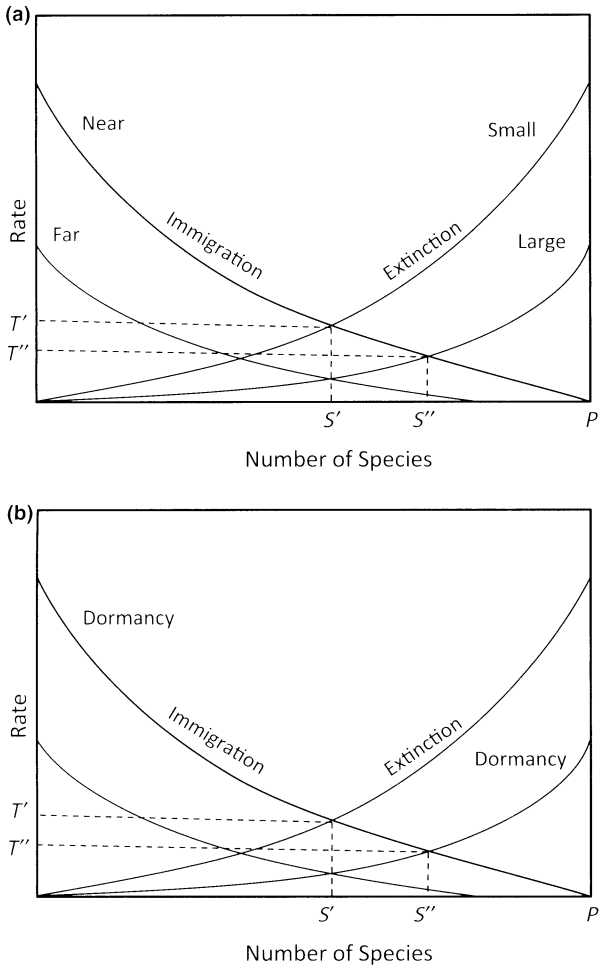

Figure 1 The classic graphical representation of the equilibrium theory of island biogeography. (a) P denotes the richness of the mainland source pool, while S' and S'' denote island species richness. T' and T'' denote temporal turnover in species richness. (b) Effects of dormancy are expected to increase the rate of immigration and decrease the rate of extinction, resulting in greater equilibrium species richness and a lower rate of temporal turnover.

INCORPORATING DORMANCY INTO THE EQUILIBRIUM THEORY

Dormancy is an alternative to extinction when growth and reproduction cannot be maintained. As more species are added to an island, more become available to go extinct or go dormant. The existence of this alternative should decrease the expected overall rate of extinction. Furthermore, as a response to resource limitations, dormancy may allow more species to persist than when all species are competing. Considering that immigrants are likely to arrive in dormant states and that they may persist if arrival conditions are adverse, continued immigration may not be accompanied by the expected difficulty of persisting on an already crowded island. Hence, dormancy is predicted to decrease the overall rate of extinction and the competition curve (Fig. 1b). However, if dormant individuals do not experience conditions suitable for reactivation, dormant populations will decrease to extinction unless maintained by immigrants – a rescue effect predicted by the equilibrium theory. Hence, extinction occurs as a stochastic process that removes small populations, including those in a state of dormancy. This necessitates the inclusion of the dormant community into the equilibrium theory and maintains extinction as a stochastic process.

The benefit of dormancy for the propagule is to extend its life and increase its tolerance for adverse conditions. Increasing the number of species capable of immigrating to an island from various sources increases the number of potential immigrants and the potential number of species encountered among immigrants. I assume that this effect translates to an increased rate of individual migration from an expanded source pool (e.g. the global community). If all species in the source pool are equally capable of persisting in a dormant state, an island would be expected to experience increased immigration and decreased extinction relative to when the source pool has a lower capacity for dormancy. Therefore, in response to an increased capacity for dormancy within the source pool, equilibrium species richness should increase towards that of the source pool, and the rate of temporal turnover should decrease (Fig. 1b). This response is similar to that expected from a large island near a mainland source where all species are potential immigrants and the probability of extinction is decreased (Fig. 1a). In addition, applying the equilibrium theory to microorganisms requires exploring the fact that, because species may be predominantly or completely dormant, if microbial communities are held in

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equilibrium at the species level, we should expect an island community at equilibrium to comprise dormant and active species.

DORMANCY AND ACTIVITY AT EQUILIBRIUM

Dormant species may outnumber active species in free-living microbial communities. For example, an additional 117 species from an initially recorded 20 species were recovered after incubation of sediment samples from Priest Pot, a pond in the English Lake District (Whitfield, 2005). From additional samples of Priest Pot and Niva Bay, a shallow-water marine bay in Denmark, 57% to 85% of the protozoan species recovered were from the rare or dormant group (Fenchel *et al.*, 1997). When dormancy is a response to resource limitation in stochastic environments, the occurrence of large dormant pools is predicted by the equilibrium theory: as species richness increases, the transition to dormancy becomes more likely and dormant individuals become less likely to reactivate. Moreover, extinctions should become more likely as dormancy becomes more common because dormant populations are incapable of maintaining a viable size without migrants. If local extinctions remain infrequent as species richness increases, the dormant community should increase in diversity as the active community decreases. This is not to imply that the dormant community should be absent at the onset of colonization, because propagules probably arrive in dormant stages.

A second, previously mentioned observation is that dormant individuals outnumber active individuals in free-living microbial systems. Unlike the previous observation, relative species abundance within and among the active and dormant communities cannot be explored with the equilibrium theory, because the equilibrium theory operates at the species level and relative abundances are unaccounted for. Hence, one drawback of the species-level approach is that we cannot account for the effects of the individual transition between dormancy and activity on relative abundance nor for the effect of relative abundance on persistence and community structure in stochastic systems. Fortunately, ecological neutral theory, an individual-based generalization of equilibrium theory, has recently been applied to microbial biogeography to address the recurring issue of the detection of rare species (Woodcock *et al.*, 2006), the importance of examining microbial community similarity (Östman *et al.*, 2009), and the importance of niche and neutral processes (Zhang *et al.*, 2009). Ecological neutral theory has also recently been modified to account for the immense sizes and diversities of microbial communities, resulting in a 'near-neutral' model based on continuous processes (Sloan *et al.*, 2006). However, ecological neutral theory has not been examined in the light of dormancy. Here, I will attempt to account for the individual transition to and from dormancy within the framework of ecological neutral theory, with the conclusion that doing so may integrate mechanisms of stabilizing (i.e. storage effect) and equalizing (i.e. fitness equivalence) coexistence.

NEUTRAL THEORY: MOVING FROM SPECIES- TO INDIVIDUAL-LEVEL RESPONSES

Ecological neutral theory has become a valuable model for community assembly, most notably since Hubbell (2001) proposed the unified neutral theory of biodiversity and biogeography (UNT). The UNT models immigration, extinction and speciation as stochastic processes that drive changes in relative abundance through dispersal limitation, disturbance and competition (Hubbell, 2001). Every individual of each species is assumed to be equivalent in their probabilities of dying, dispersing and reproducing as a result of disturbance removing individuals and creating opportunities within a resource-limited community (i.e. finite size, fully occupied) operating via one-for-one replacement (i.e. zero-sum dynamics). From these simplifying assumptions, the UNT has shown that ecologically similar species can coexist in distributions similar to those found in nature. Interestingly, assumptions of the UNT that have been criticized as unrealistic (e.g. asexual reproduction) may well characterize microbial communities, while previously unchallenged assumptions (e.g. dispersal limitation) may become the new focus of argument.

The UNT can serve as a tool for null-modelling when the effects of species differences and interactions on community assembly are largely unknown (Volkov *et al.*, 2003; Hubbell, 2006; McGill *et al.*, 2006). When UNT and non-neutral models (e.g. niche assembly theories, competitive asymmetry theories) agree with observed patterns, the burden of proof is placed on non-neutral models to explain how species differences and interactions are important at the level of community assembly and coexistence (Hubbell, 2006). This approach may prove useful when asking whether understanding patterns of microbial diversity requires accounting for species idiosyncrasies and interactions or whether a simpler set of assumptions is sufficient. We can also ask whether it is necessary to modify UNT for microorganisms, and, if so, how modifications might change UNT predictions. To initiate a discussion of how incorporating dormancy into UNT could be achieved, I will begin by adapting assumptions of dormancy from the species-based equilibrium theory to the individual-based UNT.

INCORPORATING DORMANCY INTO UNIFIED NEUTRAL THEORY

Modelling dormancy as a per capita equivalent response divides a community into an active and a dormant pool, and each species into a dormant population that experiences no births and an active population that experiences few deaths (Fig. 2). As with the equilibrium theory, assume that dormancy is an effective but temporary alternative to death. Hence, as a simplifying assumption, we could assume that the active pool experiences a negligible rate of death, and that each individual becomes metabolically inactive before dying. While this may be difficult for traditional biogeographers to accept, and while active microorganisms are likely to experience death

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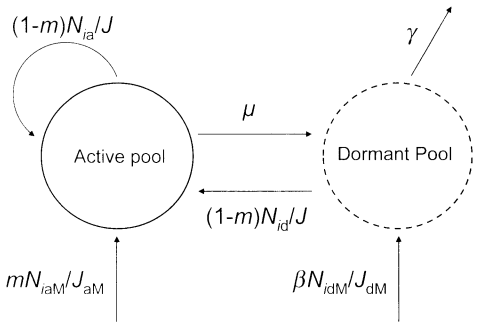

Figure 2 A conceptual representation of the unified neutral theory at the local community level, generalized to account for dormancy. Local community J is composed of active and dormant pools. Each species ($N_{i=1}...N_n$) can occur in both pools (e.g. N_{ia}, N_{id}). Disturbance forces individuals to go dormant at the rate μ, creating competitive opportunities for N_{id} to reactivate and N_{ia} to reproduce based on their relative abundances in J. The dashed circle indicates that the dormant pool is not limited in size by resources, but is checked by a death rate γ. Migration into J occurs with N_{ia} and N_{id} immigrating into their respective pools according to their relative abundances within the active and dormant pools of the metacommunity J_M. Active propagules compete with probability m that a disturbance in the active pool will result in the immigration of one individual from the metacommunity. Probability $1-m$ then represents the probability that a disturbance in the active pool will result in the reproduction or reactivation of a resident. Immigration into the dormant community is not competitive, and occurs with a constant rain β of dormant propagules. For simplicity, we could assume that one individual goes dormant and one dormant individual immigrates into J at each time step, $\mu = \beta = 1$. The ability of active individuals to produce dormant individuals or the occurrence of death without first becoming dormant could also be included, but has been avoided for simplicity. In addition, the production of dormant individuals by active individuals may be implicitly captured by the occurrence of immigration into the dormant pool.

from predation and lysis, the distinction between death and dormancy has been a source of argument in microbiology for decades (Roszak & Colwell, 1987; Colwell, 2009). In fact, natural microbial death probably ensues after the exhaustion of endogenous resources in dormant states (Colwell, 2000). The frequency of death in activity within natural environments will affect whether death in activity should be explicitly accounted for. Hereafter, death in activity is not considered, for the sake of constructing a simple conceptual generalization of the UNT.

In the UNT, disturbance removes individuals, creating opportunities for immigration and reproduction into a resource-limited community. As with the previous treatment of the equilibrium theory, allowing disturbance to instead drive individuals to dormancy creates opportunities for reactivation or reproduction into a resource-limited active pool. Immigration into the active pool is therefore competitive, as in the UNT of Hubbell (2001). The dormant pool is non-competitive because dormant individuals require few, if any, resources. Hence, the dormant pool is open and checked, but not bound in size, by death (Fig. 2). With respect to these competitive differences, immigration of dormant propagules should be relatively common and perhaps lead to a large pool of dormant individuals and rare species, predictions previously noted as generally supported for microbial communities. Transition probabilities of change in relative abundance can then be formulated from this conceptual model to examine distributions of diversity and abundance at the level of local community dynamics, although I will not present them here. As modelled here, the probability of dying in dormancy (Fig. 2) represents a probability of not persisting in a general state of inactivity. As the probability increases the model reduces to the form of Hubbell (2001). Hence, incorporating dormancy does not create a version the UNT specifically modified for species with dormant life stages, but generalizes the UNT for species with and without dormant life stages.

Generalizing the UNT to account for dormancy in an attempt to make the UNT more relevant to microorganisms is interesting for reasons unrelated to the potential success of the oversimplified model presented above. First, considering that the effects of a biological trait have not been explicitly accounted for in the UNT, the model presented here represents the first attempt to do so. Second, a model that incorporates dormancy may radically change how the UNT operates. For instance, although the active community is fixed and operates via zero-sum dynamics, the total community size is not fixed and will continue to grow indefinitely unless the consequences of dormancy prevent it. Third, until now, the UNT has lacked a mechanism to account for the common occurrence of a dormant life stage. For example, although the UNT has often been applied to tree communities, it has not been used to account for the effects of seed banks (e.g. Hubbell, 2001). Seed banks create a storage effect among plant populations by preventing local extinction through the delay of growth and reproduction until environmental conditions are favourable (Facelli *et al.*, 2005). However, the lack of a storage stage in the UNT is not surprising given that ecological neutral theory and the storage hypothesis are commonly perceived as competing hypotheses (Chesson, 2000; Hubbell, 2001), one of coexistence through fitness equivalence in which ecological species differences are ignored (neutral theory) and one of coexistence through stabilizing mechanisms in which ecological species differences are invoked (storage hypothesis). Here, the introduction of dormancy into the UNT is a first attempt to integrate individual-based storage into ecological neutral theory. The integration of equalizing and stabilizing mechanisms of coexistence has been encouraged by community ecologists and microbial ecologists who recognize the importance of niche and neutral processes (Chesson, 2000; Zhang *et al.*, 2009).

CONCLUSIONS

When accounting for the effects of dormancy on the equilibrium theory, graphical predictions of species richness and the expected proportions of dormant and active species can

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theoretically approach the predictions of the BBH. Because dormancy reduces the capacity to maintain or increase population size, local extinction may result from stochastic effects on small dormant populations. In this sense, the BBH becomes a hypothesis of dispersal assembly, wherein a resource-limited environment drives the transition to dormancy. Assuming that dispersal is affected by relative abundance and that most microorganisms in natural environments are dormant, one can envision negative effects of dormancy on dispersal. Here, I have made a first attempt to generalize conceptually the UNT to model the effects of dormancy on relative abundance and dispersal, and have concluded that doing so may integrate mechanisms of stabilizing and equalizing coexistence. Recasting the BBH in terms of dispersal and extinction and generalizing biogeographical theories to account for dormancy is a step in a new direction towards ascertaining whether microorganisms warrant a separate biogeography; similar steps should be taken to account for horizontal gene transfer and the more complex taxonomy of microorganisms.

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BIOSKETCH

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